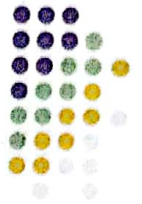


CO.
Alayen Iraqi University
Engineering College/ AI Engineering Department



3rd year

اساتذہ وزلم قلم اے
دیکھو سالانہ

Signals & Systems Lab

Lab (9)

Convolution

Lecturer: Amjed Jumaah



Discrete-Time Convolution



An **LTI** discrete-time system is (equivalently to the continuous-time case) **completely** described by its **impulse response**, which is usually denoted by **$h[n]$** .

$$y[n] = x[n] * h[n] = \sum_{k=-\infty}^{+\infty} x[k]h[n-k] = \sum_{k=-\infty}^{+\infty} h[k]x[n-k].$$

Convolution sum.

If the **unit impulse sequence** **$\delta[n]$** is the **input** to a system the **output** of the system is the **impulse response** **$h[n]$** of the system.

The Unit Impulse Sequence as Input to a System

Example

Suppose that a discrete-time system is described by the impulse response $h[n] = [2 \ 4 \ 1 \ 3]$, $0 \leq n \leq 3$. Compute the response of the system to the input signal

$$x[n] = \delta[n] = \begin{cases} 1, & n = 0 \\ 0, & n \neq 0 \end{cases}$$

The exact relationship is again $\text{length}(y) = \text{length}(x) + \text{length}(h) - 1$. Therefore, the output of the system is plotted in the **double time** interval from the input and impulse response signals.

More generally, the response of a linear shift-invariant system to a shifted unit impulse sequence $\delta[n - k]$ is shifted by k units version of its impulse response, i.e., the output of the system is $h[n - k]$.

Computation of Discrete-Time Convolution

Step 1: The **two** signals are plotted at the **k-axis**

Step 2: The reflected version of $h[k]$, namely, $h[-k]$ is plotted.

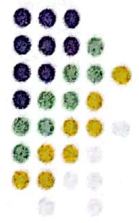
Step 3: The signal $h[n - k]$ is shifted from $-\infty$ to $+\infty$ by changing appropriately the value of n .

First stage: Zero overlap.

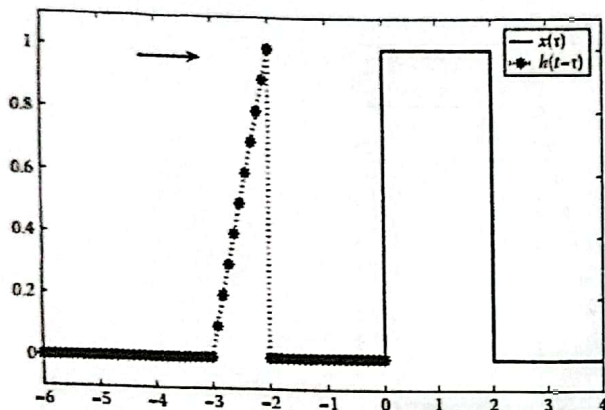
Second stage: Overlap.

Third stage: Zero overlap.

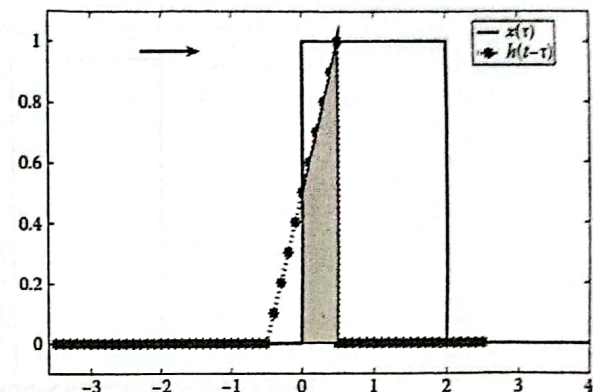
4.2.1 Computation of Convolution



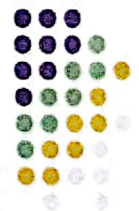
First stage: Zero overlap



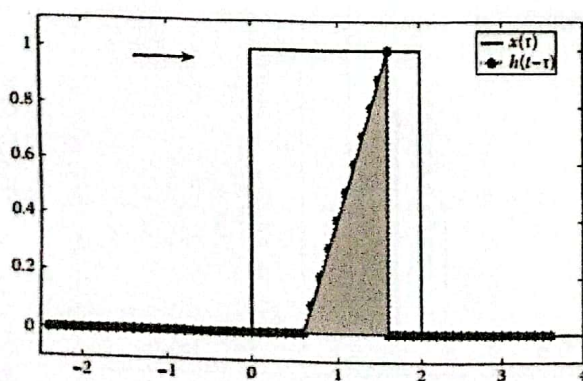
Second stage: Partial overlap



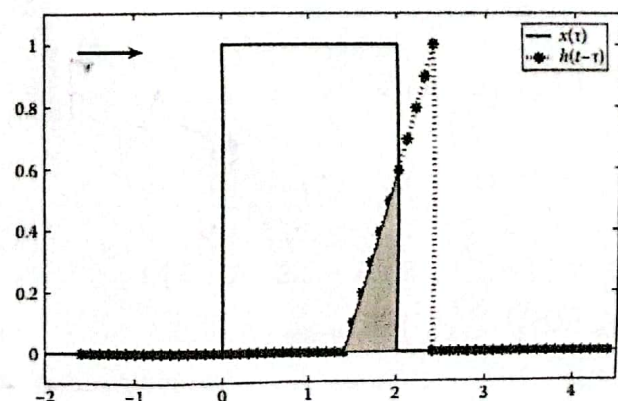
Computation of Convolution



Third stage: Complete overlap



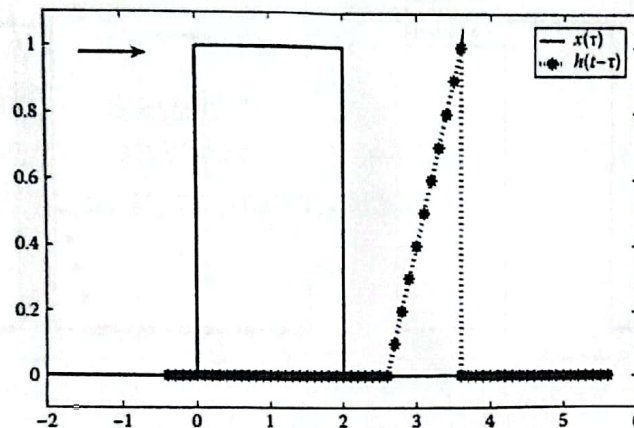
Fourth stage: Exit—partial overlap





4.2.1 Computation of Convolution

Fifth stage: Zero overlap



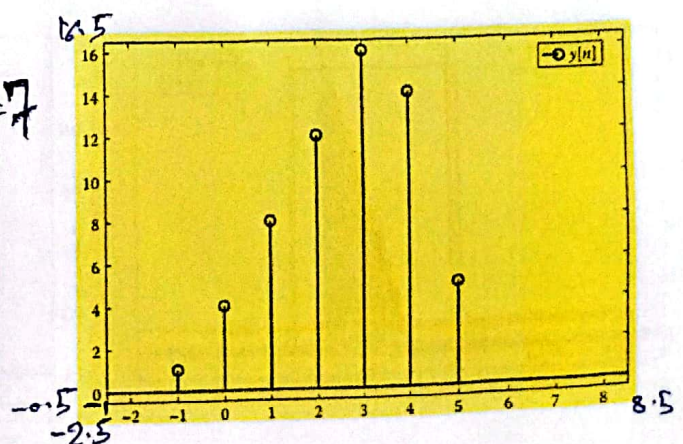
Discrete-Time Convolution Examples

Example

Suppose that the impulse response of a system is $h[n] = [1, 2, 1]$, $-1 \leq n \leq 1$. Compute the response of the system to the input signal $x[n] = [1, 2, 3, 4, 5]$, $0 \leq n \leq 4$.

```
x = [1, 2, 3, 4, 5];
h = [1, 2, 1];
y = conv(x, h);
stem(-1:5, y)
axis([-2.5 8.5 -0.5 16.5]);
legend('y[n]')
```

Handwritten notes: $8-1=7$ and arrows pointing to the x and y axes of the plot.



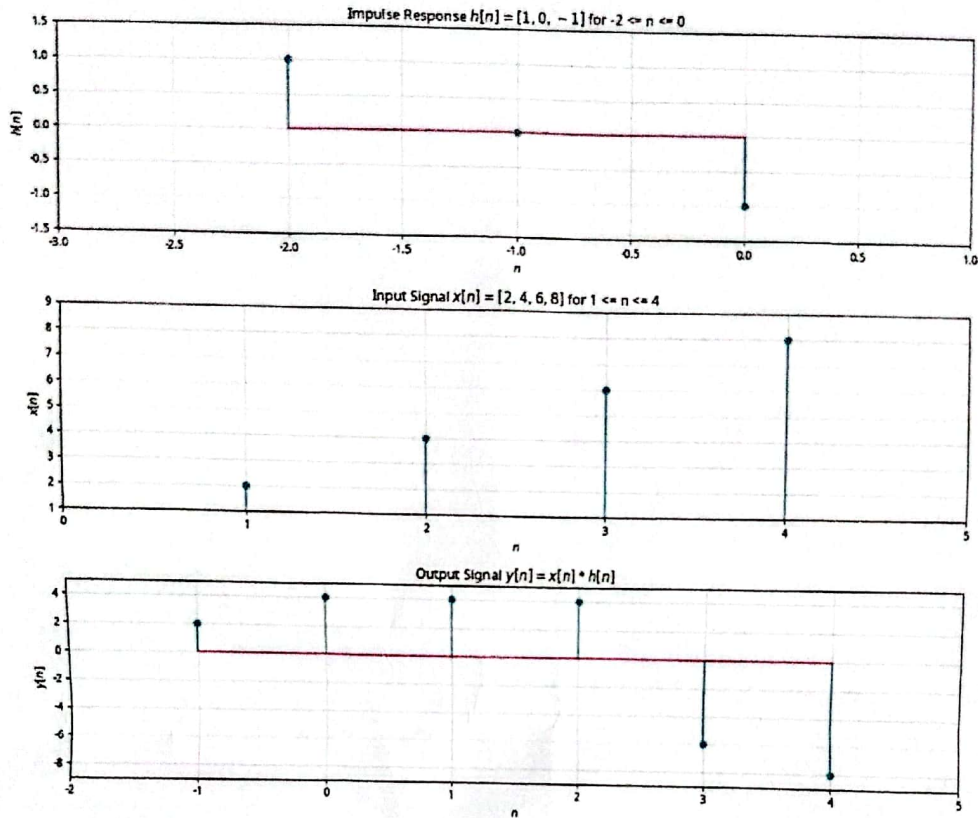
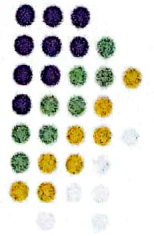
Discrete-Time Convolution Examples

Exercise

Suppose that the impulse response of a system is

Compute the response of the system to the input signal

Assignment



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